

Taehoon Kim

jerry38250325@gmail.com | [Personal Website](#) | [Google Scholar](#)

Research Interests

My research aims to develop robust perception for real-world, safety-critical environments through multimodal representation learning with computer vision and vision–language models. I am particularly interested in spatial intelligence and autonomous-driving perception via multi-sensor fusion (RGB, LiDAR, Radar). I also study industrial and multimodal anomaly detection via distillation-based teacher–student frameworks.

Keywords: Computer Vision, Spatial Intelligence, Multi-Sensor Fusion, Anomaly Detection, Vision-Language Model

Education

Kyonggi University

M.S. in Computer Science

- Advisor: Junho Ahn
- GPA: 4.5 / 4.5

Sep. 2025 – Present

Suwon, Korea

Kyonggi University

B.S. in Department of Restaurant and Culinary

B.S. in Department of Artificial Intelligence – *double major*

- CGPA: 4.3 / 4.5 – Summa Cum Laude, AI major CGPA: 4.32 / 4.5

Mar. 2019 – Aug. 2025

Seoul, Korea

Suwon, Korea

Experiences

K-SW Square, Purdue University

Visiting Research Intern

Advisor: Eric T. Matson & Anthony Smith

Sep. 2023 – Dec. 2023

West Lafayette, IN, USA

AI Vision Lab, Kyonggi University

Undergraduate Research Assistant

Advisor: Prof. Junho Ahn

May. 2024 – Aug. 2025

Suwon, Korea

Military–Academia Joint AI Education Program, ROK DCC

Teaching Assistant & Project Mentor

2024 – 2025

Suwon, Korea

National Center of Excellence in Software, Kyonggi University

Graduate Teaching Assistant

Sep. 2025 – Present

Suwon, Korea

Publications

International Journal

- [1] **Taehoon Kim**, Sehun Lee, Namgi Kim, Youjin Seo, “XSANet: Industrial X-ray Anomaly Detection via Spatial Transformer and Attention-Enhanced Teacher-Student Framework,” *Image and Vision Computing (IMAVIS)*, Elsevier, *under review*
- [2] **Taehoon Kim**, Sehun Lee, Junho Ahn, “Zero-Shot Vision-Based Robust 3D Map Reconstruction and Obstacle Detection in Geometry-Deficient Room-Scale Environments,” *Computers, Materials & Continua*, vol. 86, no. 2, pp. 1–30, Dec. 2025.
- [3] Sehun Lee, **Taehoon Kim**, Sookyun Kim, Junho Ahn, Namgi Kim, “Unsupervised Structural Defect Classification via Real-Time and Noise-Robust Method in Smartphone Small Modules,” *Electronics (MDPI)*, vol. 14, no. 17, art. no. 3455, Aug. 2025.
- [4] Sehun Lee, **Taehoon Kim**, Junho Ahn, “Zero-Shot Based Spatial AI Algorithm for Up-to-Date 3D Vision Map Generations in Highly Complex Indoor Environments,” *Computers, Materials & Continua*, vol. 84, no. 2, pp. 3623–3648, Jul. 2025.

International Conference

- [1] Doyoung Koh, **Taehoon Kim**, Sehun Lee, Namgi Kim, “Weighted Knowledge Distillation with DisturbLabel for Robust Learning,” 2026 17th International Conference on Ubiquitous and Future Networks (ICUFN), Milan, Italy, Jul. 2026.

Domestic Journal

- [1] **Taehoon Kim**, Jiyoung Na, Jiwon Yoon, Sehun Lee, Junho Ahn, “Multiple Camera-Based Real-Time Long Queue Vision

Algorithm for Public Safety and Efficiency,” Journal of the Korea Society of Computer and Information (JKSCI), vol. 29, no. 10, pp. 47–57, Oct. 2024. DOI: 10.9708/jksci.2024.29.10.047

Domestic Conference

- [1] **Taehoon Kim**, Sehun Lee, Junho Ahn, “Industrial X-ray UniNet: A Robust Spatial Transformation Architecture for X-ray Anomaly Detection in Smartphone Actuator Modules,” Proceedings of the 2025 KAD Winter Conference (**Outstanding Paper Award**)
- [2] Sehun Lee, **Taehoon Kim**, Junho Ahn, “Genetic Neural Networks: Efficient Training Initialization Strategy via Weight Recombination,” Proceedings of the 2025 KAD Winter Conference (**Best Paper Award**)
- [3] **Taehoon Kim**, Sehun Lee, SeungBeom Lee, Junho Ahn, “0-VISTA: Zero-Shot Spatial AI for 3D Mapping and Detection in Complex Indoor Environments,” Proceedings of the 2025 KAD Spring Conference
- [4] Sehun Lee, **Taehoon Kim**, Doyeon Kim, Junho Ahn, “Latent Hotspot Crossover for Disentangled and Interpretable Representation Learning in Variational Autoencoders,” Proceedings of the 2025 KAD Spring Conference (**Outstanding Paper Award**)
- [5] Doyeon Kim, SeungBeom Lee, **Taehoon Kim**, Junho Ahn, “Detection of Movable Object Layout Changes in Shared-Use Facilities,” Proceedings of the 2025 KAD Spring Conference
- [6] **Taehoon Kim**, Jiyoung Na, Sehun Lee, Jiwon Yoon, Junho Ahn, “Deep Learning Vision-Based Line Recognition Algorithm Suggestion for Real-Time Intercity Bus Passenger Queue Efficiency,” Proceedings of the 2024 KSII Spring Conference
- [7] Yechan Lee, **Taehoon Kim**, Jiyoung Na, Jiwon Yoon, Sehun Lee, Hanyong Lee, “Analysis of Autonomous Driving Algorithms for Road Driving of Delivery Robots,” Proceedings of the 2023 KIIT Summer Conference

Patent

Deep Learning-Based Queue Recognition System	Oct. 2024
Korea Patent Application	No. 10-2024-0144283

Projects

Pedestrian Safety System for Large-Scale Buildings Using 3D Congestion Mapping with Collaborative Robots and Vision AI for Disaster Response NRF	Sep. 2025 – Aug. 2026
Acoustic-Based Real-Time Drone Localization & Tracking AI System ITSA & ROK DCC	Jul. 2025 – Dec. 2025
Improvement of Detection and Classification Capabilities in SUB2 ASS’Y GAP Inspection im co., ltd.	Nov. 2024 – Feb. 2025
Real-Time Multimodal Vision-Integrated Military Fitness Assessment ITSA & ROK DCC	Sep. 2024 – Dec. 2024
Improvement of Detection and Classification Capabilities in X-RAY Ball Inspection im co., ltd.	Jul. 2024 – Aug. 2024
Enhancing Culinary Experience: Real-Time Interactive Cooking Assistance Utilizing Mixed Reality Purdue University	Sep. 2023 – Dec. 2023

Awards

Best Paper Award, 2025 Korea Data Science Winter Conference	Dec. 2025
Outstanding Paper Award, 2025 Korea Data Science Winter Conference	Dec. 2025
Outstanding Paper Award, 2025 Korea Data Science Spring Conference	Jun. 2025

Merit Award , Kyonggi University SW Imagination Company Project Competition	Nov. 2024
Gold Award , Kyonggi University Advanced Capstone Design Project Competition	Jun. 2024
Gold Award , Kyonggi University Foundational Capstone Design Project Competition	Jun. 2023
Grand Award , Kyonggi University GamsungSW IT Competency Competition	Dec. 2019

Honors

Letter of Appreciation , ROK Defense Communication Command · Award for outstanding contribution in AI development and mentoring	Apr. 2026
AI & SW Talent Scholarship , Noh Youngbaek Scholarship Foundation · Award for excellence in the AI & Software field	Nov. 2025
Faculty Council Honors Scholarship , Kyonggi University · Award for outstanding academic achievement	Jun. 2025
Highest Honors , Kyonggi University · Graduate 1st in Dept. of Restaurant & Culinary	2025
Semester High Honors , Kyonggi University · Award for perfect GPA of 4.5 / 4.5	2024
Semester High Honors , Kyonggi University · Award for Academic Excellence (1st)	2019, 2022 · 2023 (1st Sem), 2024

Technical Skills

Programming languages

Python, C, Java, R

Deep Learning Framework

PyTorch, TensorFlow, Keras, NumPy, CUDA

Language Skills

TOEIC – Total Score 920	Aug. 2024
TOEIC Writing – Advanced Mid (AM)	Apr. 2024
TOEIC Speaking – Intermediate High (IH)	Apr. 2024
Official English Presentation as University Representative Kyonggi University National Center of Excellence in Software · Presented program highlights and achievements to a CHED delegation, Philippine President's office	Jul. 2024 news article
English Interpretation & Translation Volunteer Purdue Military Research Institute (PMRI) & IITP · Interpreted AI lectures and supported communication for ROK ROTC officers	Oct. 2023